Architectural Particular Specification

PAINTING

SECTION 8.9

Rev	Date	Description
0	14 APR 2022	First Issue

SECTION 8.3

PAINTING

PART 1 GENERAL

1.01 SUMMARY OF WORKS

A. The Works of this Section includes, but is not limited to, the furnishing and installing of the following:

1. Surface preparation and field application of paints and finishes to interior and exterior surfaces and designated items.

2. Primer, paint, wood varnish, stains and other finishes.

3. Supply, deliver, apply and warrant painted finishes which shall be provided for, but not limited to, the following items. Refer to the Drawing for designated paint finishes.

   a. Exterior: All exterior surfaces including, but not limited to:

      1) Steel door frames (where not pre-finished).
      2) Metal flashing, and coping (where not pre-finished).
      3) Decorative reveals and trims (where not pre-finished)
      4) Exterior Signage(where not pre-finished)
      5) Pipe bollards and handrails
      6) Steel tubes and miscellaneous exterior metals. Access ladders.
      7) Other items and surfaces noted on the Drawings
      8) Metal roof accessories such as roof hatches, chiller screen metalwork, etc.
      9) Exposed roof mechanical equipments, ducts, etc.

   b. Interior: All interior surfaces including, but not limited to:

      1) Steel and wood door frames.
      2) Gypsum board and plaster render surfaces
      3) Coiling doors.
      4) Metal pipe bollards, stair railings and handrails
      5) Steel pan stairs & steel stairs.
      6) Exposed mechanical ductwork.
      7) Miscellaneous interior metals, as indicated.
      8) Internal wood joinery and trims as indicated.
      9) Ceiling trims and blank-out panels
      10) Other items and surfaces noted on the Drawings.
      11) Decorative steel work.

4. Non-Painted Items: Painted finishes shall not be provided for the following Items only:

   a. Aluminium, brass, bronze, stainless steel, and chrome plated steel.

   b. Pre-finished items, such as metal panels and trims, decorative wood veneer panel materials.

   c. UL, FM, and other fire certificate labels.

   d. Equipment identification, performance rating, and name plates.

   e. Finish ironmongery and hardware, U.N.O.

1.02 REFERENCE

A. Hong Kong Statutory Documents, Standards and Code of Practices.

1. Air Pollution Control (Volatile Organic Compounds) Regulation

B. British Standards:

1. BS 544 : 1969 - Specification for Linseed Oil Putty for Use in Wooden Frames.
2. BS 3761 : 1986 - Specification for Solvent-Based Paint Remover.
3. BS 3900 : All Parts - Methods of Test for Paints (date varies)
4. BS 4756 : 1998 - Specification for Ready Mixed Aluminium Priming Paints for Woodwork.
5. BS 4764: 1986 - Specification for Powder Cement Paints
6. BS 5082 : 1993 - Specification for Water-Borne Priming Paints for Woodwork.
7. BS 5358 : 1993 - Specification for Solvent-Borne Priming Paints for Wood Work.
8. BS 5493: 1977 - Code of practice for protective coating of iron and steel structures against corrosion
9. BS 6150 : 1982 - Code of Practice for Painting of Building.
10. BS 6782 : Part 1 : 1982 (1992) - Binders for Paint : Method of Determination of Volatile and Non-Volatile Content.
11. BS 7079 : All Parts - Preparation of Steel Substrates Before Application of Paints and Related Products.
12. BS 7557 : 1992 - Specification for Limits of Metal Release from Painted Surfaces of Articles, Liable to Come into Contact with Foodstuffs.
13. BS 8000 : Part 12 : 1989 - Code of Practice for Decorative Wall-Coverings and Painting.
14. Hong Kong Labour Department: A Reference Note on Ventilation & Maintenance of Ventilation Systems, June 1993.

C. American Standards

1. ASTM D660-93 : Test Method for Evaluating Degree of Checking of Exterior Paints.
2. ASTM D661-93 : Test Method for Evaluating Degree of Cracking of Exterior Paints.
3. ASTM D662-93 : Test Method for Evaluating Degree of Erosion of Exterior Paints.
4. ASTM D772-86 : Method of Evaluating Degree of Faking (scaling) of Exterior Paints.
5. ASTM D870-92 : Practice for Testing Water Resistance of Coatings Using Water Immersion.
6. ASTM D1736-89 : Test Method for Efflorescence of Interior Wall Paints.
7. ASTM D2486-89 : Test Method for Scrub Resistance of Interior Latex Flat Wall Paints.
8. ASTM D2805-88 : Test Method for Hiding Power of Paints By Reflectometry.
9. ASTM D3359-93 : Test Methods for Measuring Paint Adhesion by Tape Test.
10. ASTM D3960-93 : Practice for Determining Volatile Organic Compound (VOC) of Paints and Related Coatings.

11. ASTM D4214-89 : Method for Evaluating Degree of Chalking of Exterior Paint Films.
12. ASTM D4610-86 : Guide for Determining the Presence of and Removing Microbial (Fungal or Algal) Growth on Paint and Related Coatings.
13. ASTM D4263-93 : Test Method for Indicating Moisture in Concrete By the Plastic Sheet Method.
14. ASTM D2805-88 : Test Method for Hiding Power of Paints By Reflectometry.
15. ASTM D4834-88 (1993) : Test Method for Detection of Lead in Paint By Direct Aspiration Atomic Absorption Spectroscopy.
16. ASTM D5280 - 92: Practice for the Evaluation of Performance Characteristics of Air Quality Measurement Methods with Linear Calibration Functions.
17. ANSI/ASHRAE 62-1989 : Ventilation for Acceptable indoor Air Quality and ANSI/ASHRAE 62a-1991 Supplement.

1.03 QUALITY ASSURANCE

A. Qualification of Coating Specialist works-contractor

1. Company specializing in the installation of the systems specified herein, with minimum of 10 years documented experience, certified and approved by the coating manufacturers specified herein.

B. Qualification of Coating Manufacturers

1. Specializing in the manufacturing of high performance, commercial grade paint coating systems and materials for at least ten years.
2. The paint coating Manufacturers shall employ a Field Technical Representative available for initial, interim and final Project inspections. The Manufacturer’s Field Technical Representative shall be experienced and knowledgeable of the coating material and its application.
3. The paint coating manufacturer shall be capable of providing field service representation during construction, approving acceptable installers, application method, and conducting final inspection of the paint coating system/assembly.
4. The manufacturer shall have the quality standards of EN ISO 9001 and EN ISO 14001;
5. Valid Agency Agreement for the distributorship of the Supplier in Hong Kong from the manufacturer must be submitted for approval;
6. The manufacturer shall supply a complete VOC certified product range which meets the strict criteria of the Swiss AgBB;

1.04 PERFORMANCE REQUIREMENTS

A. Read this section in conjunction with ASD GS 2012 Section 21.

B. General

1. All materials, paints, paint products and accessories shall be new, unused and not expired.
2. All paints (including sealer, water repellent agent, and the like) for outdoor areas and all spray paint system(s) shall be proprietary product, have good adhesion and wide variety of possible textures and good spraying properties, tough and durable, and easy to maintain.
3. The entire painting system(s) including materials, installation, workmanship and performance shall be in proper and normal function in accordance with the contract requirements for a period of not less than five years from the date of substantial completion of the contract works as certified by the Architect.
4. For consistency of performance and appearance, each layer/coating of the same colour/texture of the paint system(s) for the same area/room shall be the same type/model/series by a single manufacturer & supplier. It should also be the same mix applied within the shortest time, unless proven impractical.
5. All layers of coating for each area / room shall be the same type compatible with each other and with the substrate. Statement of compatibility and suitability for the intended use of the area, from the manufacture shall be submitted by the Contractor to Architect prior to Architect’s approval of the proposed material and Contractor’s ordering. Where surfaces have been treated with preservatives or fire retardants, the layer coating materials to be compatible with the treatment and not inhibiting its performance.
6. No variation of final surface finish shall be allowed.
7. All paints to be anti-mould and stable in humid conditions and suitable to hot climate exposure.
8. Visual requirements to be based upon samples submitted and agreed and per definitions contained in BS 2015 and BS EN 971: Part 1for adhesion, excess fading, non-uniformity of colour, cracking, peeling, pitting or other visual defects. None of the defects to be acceptable. The work shall be viewed with normal eyesight from a distance of 2m for colour consistency and excessive fading and 1m for cracking, pitting and other defects. Good standards of cosmetic finish generally, when viewed from a distance of 5m or more. Minor orange peel or other texture is usually acceptable. High decorative finish shall have a high standard of evenness, smoothness and gloss when viewed from a distance of 2m or more.
9. Finishing paint of the same type and colour on same substrate at the same room or location (e.g. painting to steelwork requiring fire resistance and that for the same without fire resistance) shall be the same and from a single supplier to ensure consistency of the appearance and performance among different structural steelwork of the same area. Manufacturer’s statement and/or other written evidence shall be provided to demonstrate that the finish paint is compatible with the different base coats of both cases (i.e., with and without fire resistance base coat).
10. Premixed plaster shall be factory-produced by weighted combination of plaster raw materials and chemicals and supplied to sites in bags. Mixing with clean water shall be required in accordance with the manufacturer’s instruction.
11. Durability: Provide only paints of durable and washable quality. Do not use paint materials that will not withstand normal washing as required to remove pencil marks, ink, ordinary soil, and similar materials without showing discoloration, loss of gloss, staining, and other damage.

12. Colours and Glosses: The Architect will select colours to be used in the various locations and types of applications and will be the sole judge of acceptability of the various glosses obtained from the materials proposed to be used in the Work.

13. The Contractor shall propose to the Architect suitable materials fulfilling the requirements under this Specification

14. The whole system to be applied to each type of finishes and locations, including the bonding slurry, repairing mortar, plaster, tile adhesives, tile grout proposed to be used by the Contractor must be from the same source of manufacturer unless otherwise approved by the Architect. The Contractor must ensure that all components of the system (viz., bonding slurry coat, plaster, tile adhesive and tile grout etc.) would be suitable for use for the specific purpose of each individual case and be compatible with one another, the substrate and the finishing material.

15. The manufacturer must have the quality standards of ISO 9001 and ISO 14001.

1.05 ENVIRONMENTAL REQUIREMENTS

A. All primers, paints, solvents and other accessory materials shall be lead free and low in volatile organic compounds (VOC) in strict conformance to BS7557, BS6782 : Part 1, and ASTM D3960.

B. Ensure surface temperature or surrounding air temperature is above 4^{o}C (40^{o}F) before apply alkyd finishes; above 7^{o}C (45^{o}F) for interior latex, and 10^{o}C (50^{o}F) for exterior latex work. Minimum for varnish and transparent finishes is 18^{o}C (65^{o}F).

C. Indoor Air Quality Control

1. The Contractor is cautioned that the Works of this Section may result in the release of volatile organic compounds (VOC) and other toxins from paints, solvents and other painting products which may be attributable to the “Sick Building Syndrome (SBD)”.

2. The Contractor shall ensure that the Project Building is entirely purged of VOC and other toxins which may be attributable to SBD prior to occupancy of Project Building. Indoor Air Quality (IAQ) monitoring and testing conforming to ASTM D 5280-92, ASTM D 5014-94, ASTM D 5197-92 and ANSI/ASHRAE 62a-1991 shall be conducted before and after purging procedure. Re-purged Project Building and continue to monitor indoor air quality until acceptable IAQ is reached in strict accordance with ANI/ASHRAE standards and the Contract Documents. All air filters and grills at air inlets and outlets shall be clean of all air pollutants, toxins and microbial contaminants. Replace air filters if cleaning is not possible. Refer to Consulting Mechanical Engineer’s Contract Documents.

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

A. The Contractor and the Paint Accessory Product Manufacturers be aware of psychometric perimeters used by the Consulting Mechanical Engineer in determining indoor temperature and relative humidity requirements. Indoor air quality requirements and purging procedures.

B. The Contractor, and the paint and Paint Accessory Products Manufacturers shall warrant that all products, materials and items are suitable, and can be maintained in the various climatic and psychometric conditions encountered without adhesion or cohesion failures, blistering, chalking, checking, fading, discoloration, organic and microbial (fungal or algal) growth and other defects.

1.07 SUBMITTALS

A. Submittals as indicated in Section 1.5.

B. The contractor shall submit the following information for the Architect's approval prior to ordering for primer, paints, and coatings:-

1. Manufacturer's catalogue and specification
2. Colour chart for preliminary selection
3. Paint sample board for approval of colour and finish.
4. Colour sample board of size no smaller than 300mm x 300mm for each and every colour specified by the architect.
5. Where required, paint effect at specified area on site.
6. Compliance with Volatile Organic Compound (V.O.C.) of EPD regulates and environmental regulations.
7. Material hazard, health, safety and handling instructions.
8. Tests reports and product data on all paints, primers, stains, varnishes, solvents, paint products, and accessories.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY, STORAGE, AND HANDLING

A. Deliver paint materials in sealed original labeled containers, bearing the Manufacturer's name, type of paint, brand name, color designation, and instructions for mixing and/or reducing, pot life/expiration date, and lot numbers and use strictly in accordance with the manufacturer's recommendations.

B. Provide adequate storage facilities. Store paint materials at minimum ambient temperature of 7°C (45°F) in well ventilated, weatherproof area.

C. Store materials on site or in the contractors' workshops where shall be cool, well ventilated, covered storage space. Label tins of paint for "External Use", "Internal Use", "Undercoating" and "Finishes" shall be labelled respectively.

D. Take precautionary measures to prevent fire hazards and spontaneous combustion.

E. Off site coated steel:

1. Masking: Apply to areas of steel not requiring application of intumescent coatings.
2. Apply to Architect for approval of method statement.
3. Handling and Erection: Use methods and devices designed to minimize damage to intumescent coatings.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

(NOT USED)

1.12 WARRANTY

A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

B. The coatings and the Productions Manufacturers (in submitting a Tender providing for materials, installation and technical services and executing the Project Work) will assume to warrant that all products, material and items are suitable, and can be maintained in the various climatic and psychometric conditions encountered without adhesion or cohesion failures, blistering, chalking, fading, discoloration, organic and microbial (fungal or algal) growth and other defects for a period of five years after the practical completion of the building.

C. Minimum ten (10) years warranty against material defects and workmanship shall be provided.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURERS/SUPPLIERS

A. Subject to compliance with the Project Contract Documents, provide paint systems from the following approved paint manufacturer/ suppliers or equivalent approved by the Architect:

1. Akzo Nobel Swire Paints Limited
   Suite 2806
   Island Place Tower
   510 King's Road
   North Point
   Hong Kong
   Tel: (852) 2823 1320 Fax: (852) 2866 3232

B. Provide all materials in a coating system from the same manufacturer. Do not intermix different manufacturer's products.

2.02 MATERIALS

A. To provide the following paint system or equivalent approved by the Architect:

1. Flat Finish Emulsion Paint For Internal Wall, Ceiling and Beam

The flat finish low VOC emulsion paint system should be

Primer	ICI Dulux Total Effect Odorless Primer A931-65968 (1coat)
Top coat	ICI Dulux Forest Breath Eco-Sense All in 1 Emulsion A699 (2coats)

i Water Based Acrylic Odorless Primer

Colour	White
Finish	Flat
Composition	Pigment: Non-lead Pigment Binder: Acrylate Copolymer Solvent: Water
Volume solids	White 30%±1
VOC content	Less than 0g/L (ready to use)
Drying Time	Single coat at standard thickness (25°C, 50% R.H.) a. 0.5-1 hour for touch dry b. 2-3 hours for recoat
Toxicity	No added lead or mercury

ii Water Based Acrylic Air Purifying Emulsion

Colour	To Architect's Approval
Finish	Flat
Composition	Pigment: Non-lead Pigment Binder: Acrylate Copolymer Solvent: Water
Volume solids	White 42%±1
VOC content	Less than 2g/L (ready to use)

Drying Time	Single coat at standard thickness (25°C, 50% R.H.) 1 hour for touch dry hours for recoat
Toxicity	No added lead or mercury

The physical properties of top coat should meet following standard

Physical Property	Test Method	Typical Value
Antibacterial Test	ISO 22196 : 2011	Reduction from initial (contact time : 24 hours) Against *Escherichia coli* ≥99.99% Against *Staphylococcus aureus* ≥99.99% Against *Pseudomonas aeruginosa* ≥99.99%
Formaldehyde Emission Test	EN ISO 16000-9	0.108 mg/m^2 at 168 hours
Fungal resistance	ASTM D 5590-00	No fungal growths, rating is 0
Wet-scrub resistance	ISO 11998	EN 13300 classification : 2
Flame spread rating	ASTM E84	Class A (over non-combustible surfaces)

2. Waterborne Epoxy Coating System

The Waterborne Epoxy Polyamide paint system should be

Primer	Dulux Pro Tru-Glaze-WB 4030 Waterborne Epoxy Primer (1coat)
Top coat	Dulux Pro Tru-Glaze-WB 4406 Waterborne Epoxy Semi-Gloss Coating (2coats)

I Waterborne Polyamide Epoxy Primer

Colour	Light Grey
Finish	Flat
Volume solids	47%
VOC content	Less than 193g/L (ready to use)
Drying Time	Single coat at standard thickness (25°C, 50% R.H.) a. 2 hour for touch dry b. 16 hours for recoat
Film Thickness	a. 105-212 microns for wet film b. 50-100 microns for dry film

The physical properties of top coat should meet following standard

Physical Property	Test Method	Typical Value
Adhesion	ASTM D 4541	Excellent

Hardness	ASTM D 3363	Good
Flexibility	ASTM D 522	Excellent
Humidity Resistance	ASTM D 4585, 700 hours	Excellent

ii Waterborne Polyamide Chemical Resistant Epoxy Paint

Colour	To Architect's Approval
Finish	Gloss
Volume solids	45%
VOC content	Less than 169g/L (ready to use)
Drying Time	Single coat at standard thickness (25°C, 50% R.H.) a. 2 hours for touch dry b. 16 hours for recoat
Film Thickness	a. 111-278 microns for wet film b. 50-125 microns for dry film

The physical properties of top coat should meet following standard

Physical Property	Test Method	Typical Value
Abrasion Resistance	ASTM D 4060, CS-17 wheels, 1000g load, 1000 cycles	180mg loss
Pencil Hardness	ASTM D 3363	B
Flexibility	ASTM D 522, Method B, 1/8"	No cracking or flaking
Impact Resistance	ASTM D 2794	>100 in-lbs
Water Resistance	ASTM D 2247, 1000 hrs	No blistering and 90% gloss retention
Chemical/Stain Resistance	ASTM D 1308	Excellent
Washability	ASTM D 6142, Fed. Std. 141C, 10,000 cycles	No wear through

5. Enamel Paint For Internal/ External Metal Surface (Spray applied)

The waterborne acrylic enamel paint system should be

Primer	Dulux Pro DEVFLEX 4020 Waterborne Primer (1coat)
Top coat	Dulux Pro High Performance Waterborne Acrylic Enamel 4216 (2coats)

i Water Based Acrylic Primer

Colour	White
Finish	Flat
Volume solids	44% ± 1%
VOC content	Less than 33g/L (ready to use)

Drying Time	Single coat at standard thickness (25°C, 50% R.H.) a. 0.5 hour for touch dry b. 2 hours for recoat
Film Thickness	125-200 microns for wet film 55-88 microns for dry film
Toxicity	Lead, chromate and mercury components free

The physical properties of top coat should meet following standard

Physical Property	Test Method	Typical Value
Pencil Hardness	ASTM D 3363	2B
Flexibility	ASTM D 522, Method B, 1/8”	No cracking or flaking
Impact Resistance	ASTM D 2794	160 in-lbs
Humiduty / Corrosion Resistance	ASTM D 4585, 1000 hours	No effect
Flash Rust Resistance	Sandblasted Steel@ 90F & 90% RH	No rusting

ii Water Based Acrylic Enamel Paint

Colour	To Architect’s Approval
Finish	Flat / Semi-gloss
Volume solids	36% ± 1% - varies with colour
VOC content	Less than 47g/L (ready to use)
Drying Time	Single coat at standard thickness (25°C, 50% R.H.) a. 1 hour for touch dry b. 4 hours for recoat
Film Thickness	106-290 microns for wet film 38-100 microns for dry film
Toxicity	Lead, chromate and mercury components free

The physical properties of top coat should meet following standard

Physical Property	Test Method	Typical Value
Abrasion Resistance	ASTM D 4060, CS-17 wheels, 1000g load, 300 cycles	40mg loss
Pencil Hardness	ASTM D 3363	2B
Flexibility	ASTM D 522, 1/8” mandrel	No cracking or flaking
Impact Resistance	ASTM D 2794	160 in-lbs
Accelerated weathering	ASTM G 23, QUV 1000 hrs	100% gloss retention
Dry Opacity	ASTM D 2805	0.994
Flame Spread Rating	ASTM E 84 05	Class A

3. Emulsion Paint For External Wall, Ceiling, Beam and Dado

The waterborne silicone external emulsion paint system should be

Primer	Dulux Pro Weathershield Primer A931-65978 (1coat)

Top coat	Dulux Pro Silicone Flexi Exterior A288 (2coats)

i Water Based Copolymer Primer

Colour	White
Composition	Pigment: Lightfast Non-Lead Pigment Binder: Styrene Acrylic Copolymer Solvent: Water
Volume solids	35% (nominal)
VOC content	Less than 47g/L (ready to use)
Drying Time	Single coat at standard thickness (25°C, 50% R.H.) a. 0.5-1 hour for touch dry b. 2-3 hours for recoat
Toxicity	No added lead or mercury

ii Waterborne silicone Paint

Colour	To Architect’s Approval
Finish	Low Sheen
Composition	Pigment: Lightfast Non-Lead Pigment Binder: Pure acrylic polymer and silicon resin Solvent: Water
Volume solids	White 45% (nominal), other colors will vary
VOC content	Less than 21g/L (ready to use)
Drying Time	Single coat at standard thickness (25°C, 50% R.H.) a. 1-2 hours for touch dry b. 4 hours for recoat
Toxicity	No added lead or mercury

The physical properties of top coat should meet following standard

Physical Property	Test Method	Typical Value
Impact Resistance	ASTM D2794-93	No cracking and no loss of adhesion was observed
Artificial weathering test for 1000 hours	ASTM G53-88	No color change and >99% of gloss retained
Water resistance	ASTM D870-97	No discernible change
Anti-carbonation	BS-EN 1062-6: 2002	R = 83m
Dirt rtention	ASTM 3719-95	Dirt collection index : 97 After 61 days exposure
Water vapour transmission	ASTM D1653-93 Method B - Wet Cup Method	220 g/m^2 per 24hrs

B. Paint Accessory Materials: Linseed oil, shellac, turpentine and other materials not specifically indicated herein, but required, to achieve the finishes specified of high quality and from approved Manufacturers.

C. Paints: Ready-mixed; pigments fully ground maintaining a soft paste consistency, capable of readily and uniformly dispersing to a complete homogeneous mixture.

D. Paints to have good flowing and brushing properties and be capable of drying or curing free of streaks or sags.

E. The following Paint Schedule provides for minimum three-coat application, unless noted otherwise. Third coat may be required as directed to provide complete coverage and uniform appearance. Omit primer for items shop primed, unless primed surfaces show signs of poor adhesion and cohesion, defects and other detrimental conditions.

F. Re-prepare and reprime such surfaces as part of the Work.

G. Primers, undercoats and finishing coats shall be appropriate for use with each other as supplied and recommended by a single manufacturer; undercoats shall also be the appropriated shade for use with the finishing coat.

H. Coating on Structural Steel: For indoor use on structural steel members like columns, girders and framework with a highly effective protection to delay the steel from reaching its critical temperature with fire resistance protection of 1-hour or 2-hour.

Note: With critical situation i.e. frequent formation of condensation and/or heating up of surfaces above 45℃, possible special measures should be taken.

Option 1

unitherm 38091 - interior
primer for steel permacor 1705
or
primer for galvanized steel permacor 2706/EG
solvent based fire protection coating
unitherm 38091 – interior use
top coat unitherm 7854
No top coat required for internal dry environments except for a coloured decorative finish

Mass density:	approx. 1,29 g/cm^3
Weight solid:	approx. 70% (according to EN ISO 3251)
VOC:	approx. 400 g/l (according to EN ISO 11890-1)
Flash point:	+26℃
IMDG-Code No.:	Class 3.3, UN-No.: 1263
Shelf life:	12 months from delivery in cool and dry storage conditions, original unopened containers
Theoretical coverage:	fire rate unitherm 38091 according BS 476 Part 21 Example: 500µm dry – 750-800µm wet* - 1000g/m^2 – 0,780 l/m^2 See corresponding separate Consumption Table. *Note: Ratio dft / wft cvaries depending on the application method.

Option 2

Sigma Coatings 4-side Protection System
1) Sigma Core 256
2) Sigma Cover S22
3) Sigma Fire Barr 100 QD
4) Sigma Vikote 56

PART 3 EXECUTION

3.01 EXAMINATION

A. Applicator shall examine areas and conditions under which painting work is to be applied and notify the Contractor in writing of conditions detrimental to proper and timely completion of work. Do not proceed with work until unsatisfactory conditions have been corrected.

B. Starting Work constitutes acceptance of the existing conditions and the Contractor shall, at his own expense, be responsible for correcting all unsatisfactory and defective Work encountered.

3.02 PREPARATION

A. Do not paint over dirt, rust, scale, grease, moisture, scuffed surfaces, and conditions otherwise detrimental to formation of a durable paint film.

B. Do not carry out works in wet humid or foggy weather, direct sunlight or on surfaces which are not thoroughly dry or if there is excess dust in the air.

C. Perform preparation and cleaning procedures in accordance with paint Manufacturer's printed instructions for each particular substrate condition.

1. Provide barrier coats over incompatible primers, or remove and reprime as required.
2. Remove finish ironmongery, hardware, hardware accessories, machined surfaces, plates, lighting fixtures, and similar items in place and not to be painted, or provide surface applied protection prior to surface preparation and painting operations. Reinstall all removed items after completion of paint work.
3. Ensure that all holes, cracks and other defects in surfaces have been made good prior to painting.
4. Clean surfaces to be painted before applying paint of surface treatment. Remove oil and grease prior to mechanical cleaning.

D. Ferrous Metals: Clean ferrous surfaces that are not galvanized or shop-coated, free of oil, grease, dirt, loose mill scale and other foreign substances by solvent or mechanical cleaning.

1. Touch-up shop-applied prime coats, where damaged or bare. Clean and touch-up with same type of shop primer.

E. Galvanized Surfaces: Clean free of oil and surface contaminants with non-petroleum based solvent. Apply coat of etching primer if required by Paint Manufacturer.

F. Cementitious Materials: Prepare cementitious surfaces of concrete, brick, and cement plaster to be painted by removing efflorescence, chalk, dust, dirt, grease, oils, and by roughening as required to remove glaze.

1. Determine alkalinity and moisture content of surfaces to be painted by performing appropriate tests.
2. If surfaces are found to be sufficiently alkaline to cause blistering and burning of finish paint, correct condition before application of paint.
3. Do not paint over surfaces where moisture content exceeds that permitted in Manufacturer's printed instructions.

4. Clean floor surfaces scheduled to be painted with a commercial solution of muriatic acid, or other etching cleaner. Flush floor with clean water to neutralize acid, and allow to dry before painting.

G. Wood: Clean wood surfaces to be painted of dirt, oil, and other foreign substances with scrapers, mineral spirits, and sandpaper, as required. Sandpaper smooth those finished surfaces exposed to view, and dust off. Scrape and clean small, dry, seasoned knots and apply a thin coat of white shellac or other recommended knot sealer, before application of priming coat. After priming, fill holes, and imperfections in finish surfaces with putty or plastic wood-filler. Sandpaper smooth when dried.

1. Prime, stain, or seal wood required to be job-painted immediately upon delivery to job. Prime edges, ends faces, undersides, and backsides of such wood, including cabinets, counters, cases, trims, skirting, and panelling.
2. Use spar varnish for backpriming transparent finish surfaces.
3. Backprime panelling on interior partitions only where masonry, plaster, or other wet wall construction occurs on backside.
4. Seal tops, bottoms, and cut-outs of unprimed wood doors with a heavy coat of varnish or equivalent sealer immediately upon delivery to Project Site.

3.03 APPLICATION

A. Do not use rollers, cloths, gloves or mechanical spraying machine unless ordered or approved by Architect. When mechanical spray painting is ordered or permitted. The priming coat shall be applied with brush. Metal Surface shall be spray applied.

B. Apply each coat at proper consistency. Protect other surfaces from paint and damage. Repair damage as a result of inadequate or unsuitable protection.

C. Each coat of paint is to be slightly darker than preceding coat unless otherwise approved by the Architect. Sand lightly between coats if necessary to achieve required finish.

D. Apply paint finishes only when moisture content of surfaces is within acceptable ranges for type of finish being applied.

E. The Paint Schedule provides for a minimum three-coat application, unless noted otherwise. The intent of these Specifications is to provide 100% protective coverage for all exposed surfaces scheduled for painting.

1. Additional coat may be required by Architect to give complete coverage and uniform appearance.
2. Primer may be omitted for items shop primed.
3. Paint inside surfaces of sheet metal items and assemblies.

F. Apply additional coats when undercoats, stains, or other conditions show through final coat of paint until paint film is of uniform finish, color, and appearance. Give special attention to ensure that surfaces, including edges, corners, crevices, welds, and exposed fasteners, receive a dry film thickness equivalent to that of flat surfaces.

G. Minimum Coating Thickness: Apply materials at not less than the Manufacturer’s recommended spreading rate. Providing a total dry film thickness (DFT) of the entire system as recommended by the Manufacturer, or as specified herein for stricter DFT.

H. Prime Coats: Before application of finish coats, apply a prime coat of material as recommended by the Manufacturer to material that is required to be painted or finished and has not been prime coated by others. Recoat primed and sealed surfaces where evidence of suction spots or unsealed areas in first coat appears, to assure a finish coat with no burn through or other defects due to insufficient sealing.

I. Pigmented (Opaque) Finishes: Completely cover to provide an opaque, smooth surface of uniform finish, color, appearance, and coverage. Cloudiness, spotting, laps, brush marks, runs, sats, or other surface imperfections will not be acceptable.

J. Transparent (Clear) Finishes: Use multiple coats to produce a glass-smooth surface film of even luster. Provide a finish free of laps, cloudiness, color irregularity, brush marks, runs, sags, orange peel, nail holes or other surface imperfections.

K. Completed Work: Match approved samples for color, texture, and coverage. Remove, refinish, or repaint all work not in compliance with specified requirements.

L. Provide adequate fresh air, exhausting and other measures to ensure that specified indoor air quality during and after Work of this Section is met. All Volatile Organic Compounds, air pollutants and other toxins released during Work of this Section shall be purged in strict accordance with the Consulting Mechanical Engineer's Contract Documents.

M. The following shall be carried out in relation to mechanical and electrical equipment:

1. Replace identification markings on mechanical, electrical and other equipments when painted over or spattered.
2. Prepaint gas piping prior to installation.
3. Paint exposed pipes, conduit and electrical equipment occurring in finished areas where it will be exposed to the public. Color and texture to match adjacent surfaces.
4. Paint both sides and edges of plywood backboards for electrical equipment and other applications before installing backboards and mounting equipment on them.
5. Refer to Consulting Engineer's Contract Documents for other painting requirements.

N. Touch up coated surfaces on completion.

3.04 CLEANING

A. As Work proceeds, and upon completion, promptly remove paint where spilled, splashed, or spattered. During progress of Work keep premises free from any unnecessary accumulation of tools, equipment, surplus materials, and debris.

B. Upon completion of Work leave premises neat and clean.

C. Legally transport and dispose of all paints, solvents and other painting products in accordance with local Environmental Protection regulations.

3.05 PROTECTION

A. Protect other surfaces from paint and damage. Repair damage as a result of inadequate or unsuitable protection.

B. Install "Wet Paint. Do not touch." signs in English and Chinese and barrier tapes around painted areas and items.

END OF SECTION